

I build systems and compiler components in C++23/C17/Rust: LLVM passes, memory-dependence analysis, CFG/SSA, graph algorithms, x86-64 code generation, and reproducible test oracles.

MCST · LLVM LICM

C++ pass, conservative safety gates, semantic before/after harness, -verify-each

1st of 49 · 104/104

Conservative PS-form memory-dependence analysis with an exact oracle

x86-64 pipeline

SSA-like IR, liveness, linear scan, iced-x86, and differential validation

EXPERIENCE

- July 1 — August 31, 2026

MCST — compiler engineering intern

 - Implemented an LLVM LICM pass in C++ with checks for side effects, memory access, loop invariance, and speculative safety.
 - Validate baseline/optimized semantic parity through Ili, CHANGE/NO_CHANGE cases, and IR verification with -verify-each.
- 2026

PS-form Memory Dependence Analyzer

 - Implemented conservative analysis of parametric memory accesses; ranked 1st of 49 with 104/104 tests.
 - Built an exact oracle for small domains, metamorphic tests, and reproducible WA/TL counterexamples.
- 2026

x86-64 Codegen & Register Allocation Lab

 - Designed SSA-like IR, liveness, linear scan, and parity checks between an IR interpreter and generated machine code.

PROJECTS

- PS-form Memory Dependence Analyzer

Ranked 1st of 49 with the only 5.0/5.0 score and 104/104 official tests; the post-selection harness preserves reproducible wrong-answer/time-limit counterexamples.
- x86-64 Codegen & Register Allocation Lab

A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.
- UniversalToolchain / Wist2

Semantic verification and backend-parity checks act as fail-closed regression gates, with verification records tied to exact revision/run identities.

SKILLS

- C++ / toolchain

C++23, C17, CMake, Linux, clang-tidy, ASan/UBSan, GitHub Actions
- Compiler analysis

CFG, dominance, SSA, SCC, dataflow, memory dependence, optimization passes
- Code generation

x86-64, iced-x86, liveness, linear scan, IR interpretation, isolated execution
- Verification

exact oracles, differential/metamorphic tests, stress testing, counterexample reduction

EDUCATION

Software Engineering student at HSE University

RECOGNITION

- The Program Committee accepted “Build the Language, Then Make the Abstractions Disappear: Extensible Programming on .NET” for presentation on UniversalToolchain/Wist2 architecture in Torremolinos, Spain.

Moscow / remote